Total No. of Printed Pages: 1

SUBJECT CODE NO: - RNP-8023
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - III
Examination November / December 2025 (To be held in February-2026)
Electronics Devices & Circuits

[Time: 2 Hours] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1) Part A: Question No 1 is compulsory.
2) Attempt any FOUR questions from Part B: Q.2 to Q.7.
3) Use of non-programmable calculator is allowed

Part A

Q1 **Attempt ANY FIVE of the following questions** **10**

a) Draw the Symbols BJT, FET & MOSFET
b) Define Load Regulation & Line Regulation
c) Draw the Drain & Transfer Characteristics of N-Channel FET"
d) Sate the Barkhusain Critera for Oscillations
e) Match the pairs:

Sr. No.	Parameters	Sr. No.	Options
1	Crystal Oscillator	a	Removes Ripple
2	Voltage Divider Circuit	b	Produces High frequency signal
3	Voltage Multiplier	c	Universal Biasing
4	Filter circuit	d	Converts low AC to high DC voltage

f) State the advantages & Disadvantages of negative feedback
g) Define oscillator with Block diagram

Part B

Q. 2 Draw & Explain the Block diagram of DC regulated Power Supply, with wave-forms **5**

Q. 3 Define a Filter and explain all types of filters with input & output waveforms. **5**

Q. 4 Draw & Explain the circuit diagram of BJT based RC coupled amplifier **5**

Q. 5 What is BJT Biasing? State the types of Biasing & explain any one type **5**

Q. 6 Define an FET. Explain working of N-Channel FET with suitable diagrams **5**

Q. 7 Define an oscillator? Explain Colpit's Oscillator in details **5**